Miss Two, Win Chicken

How the upcoming-game predictions work · November 5, 2026 · data through November 2, 2026

In short

- **What we predict.** For each upcoming game, the chance that one visiting player misses both free throws of a trip to the line in the 4th quarter, the moment the crowd wins a free chicken sandwich.
- **Data.** Every play of 3,797 NBA regular-season games since October 2023, including 48,233 fourth-quarter free throws, plus the full 2026-27 schedule.
- **Model.** For each visiting player who has been playing, multiply how often he goes to the line late by his chance of missing twice. Add that up across the roster, scale it by how often the home team fouls late, and convert the total to a probability using past seasons.
- **How good.** Better than a team's track record or the league average, but modest. In a full held-out season, the games it rated highest hit 20.8% of the time and those it rated lowest 11.4%. Treat it as a lean, not a lock.
- **Right now.** An average remaining game sits at 16%. The best chance in the next week is Lakers at Bucks on Nov 11, at 26%.

1. What we are predicting

The promo pays out when **one visiting player misses two free throws in a row on the same trip to the line in the 4th quarter**. That means both shots of a two-shot trip, or two straight misses on a three-shot trip. A teammate's technical free throw in the middle of the trip does not break it. Overtime does not count, and neither do single free throws such as and-ones.

For every game the outcome is simply yes or no. The model's job is to put a probability on the yes, before the game starts. Here is how often it has happened:

Season	Road games	Promo nights	Share of games	Per 41 home games
2023-24	1,226	186	15.2%	6.2
2024-25	1,225	166	13.6%	5.6
2025-26	1,225	217	17.7%	7.3
2026-27 so far	107	27	25.2%	10.3

So the starting point for any game is roughly a one-in-six chance. The early 2026-27 rate is based on only 107 games and will likely settle back toward the long-run level.

2. The data

Source	Seasons	What it provides
NBA live play-by-play feed, archived on GitHub by shufinskiy/nba_data	2023-24, 2024-25, 2025-26	Every play, including each free throw: shooter, team, quarter, clock, trip position ("1 of 2", "2 of 2") and make or miss.
ESPN play-by-play and schedule, archived daily on GitHub by sportsdataverse	2025-26, 2026-27 so far, remaining schedule	The same play detail for the current season, plus date, tip time and arena for every game not yet played.

- **Volume:** 3,797 regular-season games, 48,233 fourth-quarter free throw attempts and 82,193 player appearances.
- **Cross-check:** both feeds cover 2025-26. On the 1,229 games they share, they agree exactly on every visitor 4th-quarter free throw, trip and promo night (217 in each).
- **Players:** the two feeds number players differently; matching their 2025-26 player lines links 562 of 563 players.
- **Left out:** 14 neutral-site games (Mexico City, Paris, Berlin, London and the NBA Cup semifinals), where no home crowd runs a promo, plus All-Star games and the NBA Cup final, which are not regular-season games.
- **Freshness:** current through Nov 2, 2026 (107 games; there were no games on Nov 3). A scheduled job refreshes the data, the predictions and the dashboard every afternoon once the archive posts the previous night's games. (NBA.com itself is blocked from the environment that runs this, so the pipeline reads the GitHub archives of the same feeds.)

3. What the history showed, and why the model is built around players

Before building anything, I checked which signals actually repeat. Over the three complete seasons:

- **Visitors miss a bit more.** Visitors set off the promo in 15.5% of games, home teams in 13.7%. Visitors also trigger it 15% more often than their shooters' usual free throw % predicts (569 vs. 495). The effect is small but real (paired test p = 0.049), and the final calibration step absorbs it.
- **Team history barely carries over.** A team's road promo rate one season says almost nothing about the next. Free throw percentage carries over much better, because shooting skill belongs to players and players are stable.
- **Arenas look like chance.** Randomly shuffling which arena each game was played in produces spreads as wide as the real ones 46% of the time, so the model has no arena term beyond the home team's late fouling.
- **The culprits are poor free throw shooters who get fouled late.** Over three seasons, Giannis Antetokounmpo (64% on free throws) was involved in 14 promo nights as a visitor, and Rudy Gobert (61%) in 9.

Measure, correlated across consecutive seasons	2023-24 to 2024-25	2024-25 to 2025-26
Visiting team's road promo rate	+0.09	-0.03
Visiting team's free throw %	+0.26	+0.29
Visitor promo rate at each arena	+0.06	+0.11

A correlation of 0 means last season tells you nothing about this one; 1 means it tells you everything.

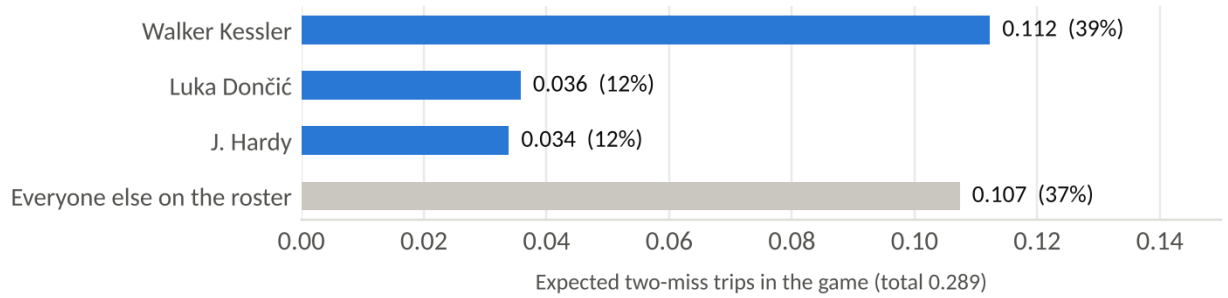
4. The model, step by step

The model estimates how many times, in expectation, a visiting player will miss both shots of a 4th-quarter trip, then turns that expected count into a probability. Six steps, all computed from games before the one being predicted:

Step	What it estimates	How
1. Who is playing	Visitors who appeared in the team's last three games	Weighted by how many of the three they played, so an injured player drops out.
2. Late trips	4th-quarter two- and three-shot trips per game	This season blended with last season (worth 10 games). Rookies start at 70% of the league rate.
3. Chance of two misses	$(1 - FT\%)^2$ for a two-shot trip	FT% from this season and last plus 20 attempts at the league average, so a 5-for-10 start does not make a 50% shooter. Three-shot trips (4% of late trips) count any two straight misses.
4. Home team's late fouling	Opponents' late trips against this team vs. the league	Pulled toward average with 15 games of weight. Milwaukee's 1.12 means 12% more late trips than average.
5. Add it up	Expected two-miss trips, λ	λ = fouling × Σ over visiting players of (availability × late trips × chance of two misses).
6. Probability	Fit on two backtest seasons	odds = 1.21 × λ, so chance = 1.21λ / (1 + 1.21λ). The 1.21 absorbs road pressure and off nights that shot-by-shot math misses.

Worked example: Lakers at Bucks, Wednesday, Nov 11, 8:00 PM ET (the week's highest chance)

Visiting player	FT% used	Expected late trips	Chance of two misses per trip	Expected two-miss trips
Walker Kessler	65.7%	0.93	12.1%	0.112
Luka Dončić	79.0%	0.79	4.5%	0.036
J. Hardy	74.9%	0.52	6.5%	0.034
Everyone else on the roster				0.107
Game total (Bucks' fouling factor 1.12 included)		5.0		0.289

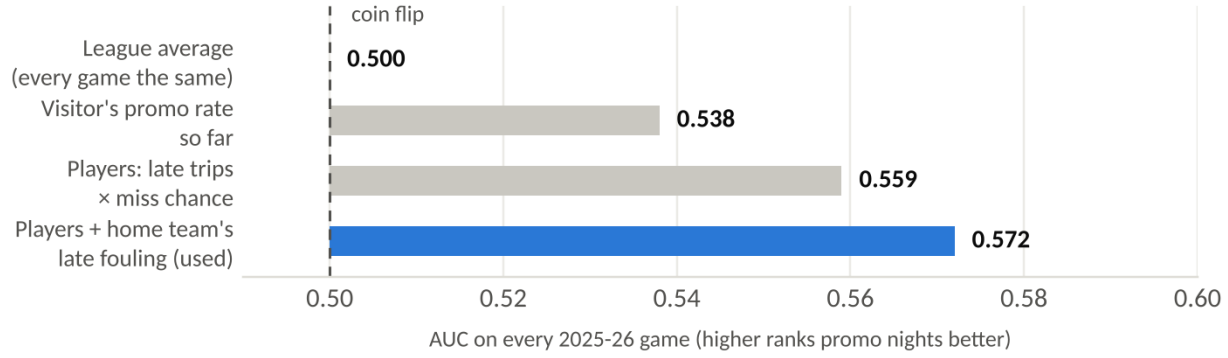


Expected late trips already include each player's availability and Milwaukee's late fouling. Walker Kessler alone carries 39% of the risk: a 66% shooter who gets to the line late almost once a game. The game total of 0.289 expected two-miss trips converts to a 25.1% chance on shot-by-shot math alone, and to **26.0%** after the calibration in step 6.

5. How well it works

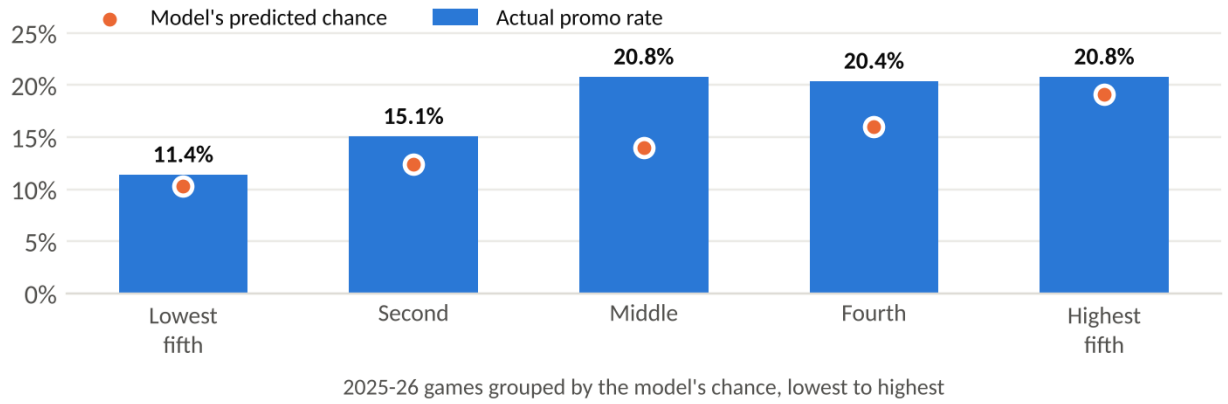
To test it fairly, the model was fit on 2024-25 and then scored every 2025-26 game (1,225 games), using only what was known before each tip-off. Nothing about the test season was used to build it. Four approaches were compared:

Model	AUC	Brier skill vs. league average
League average	0.500	0.0%
Visitor's promo rate so far	0.538	0.5%
Players: trips x miss chance	0.559	0.5%
Players + home team's late fouling (the one used)	0.572	1.1%



AUC is the chance the model rates a random promo night above a random non-promo night: 0.5 is a coin flip, 1.0 is perfect. **Brier skill** is how much the probabilities' squared error improves on giving every game the league average.

Predicted vs. actual, by fifth of games

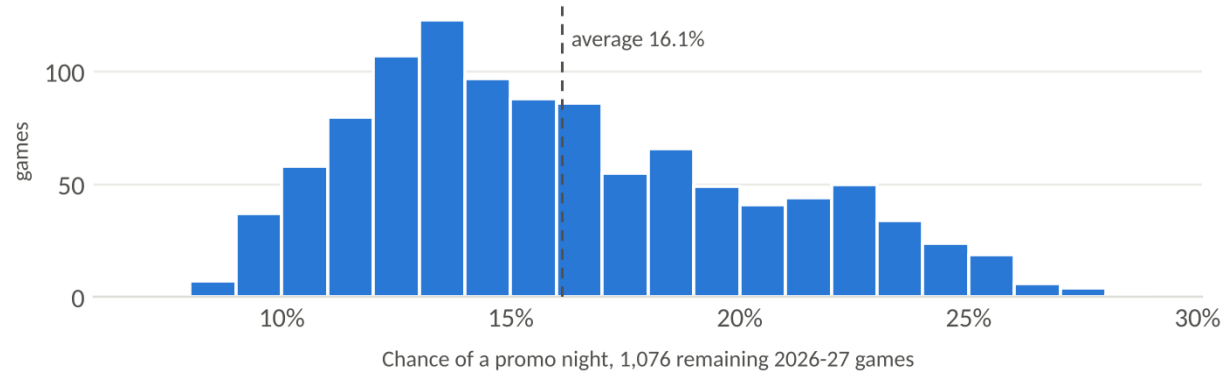


2025-26 games grouped by the model's chance, lowest to highest

The games the model rated lowest came in at 11.4%; the other four fifths ran 15-21%. It is better at spotting unlikely nights than at singling out sure things: the middle, fourth and top fifths all landed near 20-21%. 2025-26 also ran hotter than the training season (17.7% vs. 13.6%), so its predictions came in low; the version used now is fit on both seasons. Scored the same way, 2024-25 gives the same AUC (0.571), and in both seasons the top-rated tenth of games hit 21-22% of the time.

6. What it says right now

As of November 5, 1,076 regular-season games remain. Their chances range from 8.4% to 28.2%, and eight in ten fall between 11.1% and 22.5%.



Chance of a promo night, 1,076 remaining 2026-27 games

Highest chances in the next seven days

Date	Game	Chance	Most likely to miss (FT%)
Wed, Nov 11	Lakers at Bucks	26.0%	Walker Kessler (66%), Luka Dončić (79%)
Tue, Nov 10	Heat at Jazz	25.9%	Giannis Antetokounmpo (65%), Donovan Mitchell (68%)
Fri, Nov 6	Timberwolves at Pelicans	24.1%	Rudy Gobert (53%), J. Kuminga (71%)
Mon, Nov 9	Lakers at Hawks	23.8%	Walker Kessler (66%), Luka Dončić (79%)
Sun, Nov 8	Heat at Kings	23.2%	Giannis Antetokounmpo (65%), Donovan Mitchell (68%)
Fri, Nov 6	Raptors at Bucks	23.1%	R. Barrett (71%), S. Barnes (80%)
Thu, Nov 5	Heat at Warriors	23.1%	Giannis Antetokounmpo (65%), Donovan Mitchell (68%)
Wed, Nov 11	Cavaliers at Knicks	22.3%	Evan Mobley (62%), J. Allen (71%)

- **Riskiest visitors on average this season:** Heat 24.6%, Lakers 23.9%, Cavaliers 22.3%, Timberwolves 22.0%, Bulls 21.6%.
- **Safest visitors:** Magic 9.8%, Thunder 10.4%, Knicks 10.7%.
- **How to read a chance:** 26% means about one such night in four. Even the best game is far more likely to pass without a sandwich than with one, so the list is most useful for comparing nights, for example to decide which home games get extra promo staff or social posts.

7. Limits, and what would make it better

- **A snapshot of who is playing.** Rosters and availability come from each team's last three games. Injuries, trades and minutes changes after the data date are not reflected until the next refresh, and games weeks away carry today's form.
- **No minutes projections or betting lines.** Close games produce more late fouling, and a pregame point spread would capture expected closeness. Neither is in the model yet.
- **Early-season noise.** With about seven games per team, current-season numbers are thin; the model leans on last season and the league average until more games come in.
- **Tested and left out:** arena effects (no repeatable signal) and team promo history (weaker than player shooting).
- **The biggest upgrade would be a live version.** The player who actually went to the line matters far more than anything known before tip-off. A model updated at the start of the 4th quarter (who is on the floor, the score, fouls to give) should be much sharper. Adding injury reports and projected minutes would sharpen the pre-game version.

Glossary

Term	Meaning
Trip to the line	One visit to the free throw line: two shots after most shooting fouls, three after a foul on a three-point attempt.

Shrinkage	Pulling a small-sample number toward a sensible average (last season, or the league) so a few games don't swing it.
Calibration	Making probabilities mean what they say: games given 20% should happen about 20% of the time.
Backtest	Scoring past games as if they hadn't happened yet, using only data from before each one.

Files and code: **nba-ft-promo/** in the site repository holds the scripts (build_dataset.py, combine.py, analyze.py, predict.py), the daily refresh (refresh.sh) and the data, including upcoming_predictions.csv for Tableau. The interactive dashboard is the Claude artifact "Miss Two, Win Chicken".